

MacSet CX1 4G LTE Cellular Relay Controller Datasheet

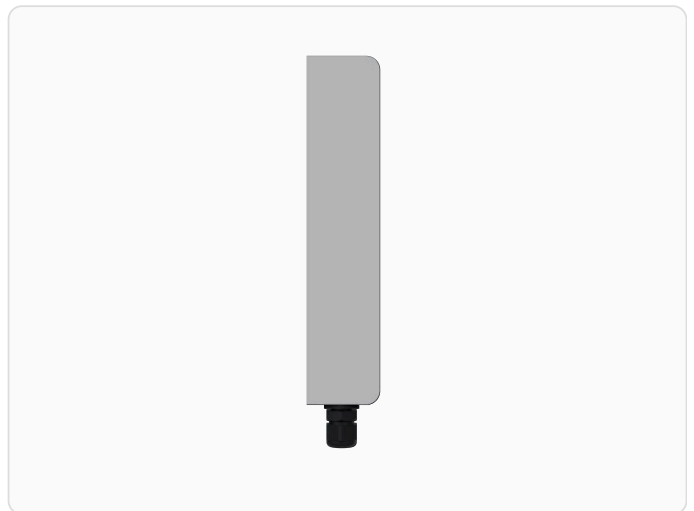
Product Overview

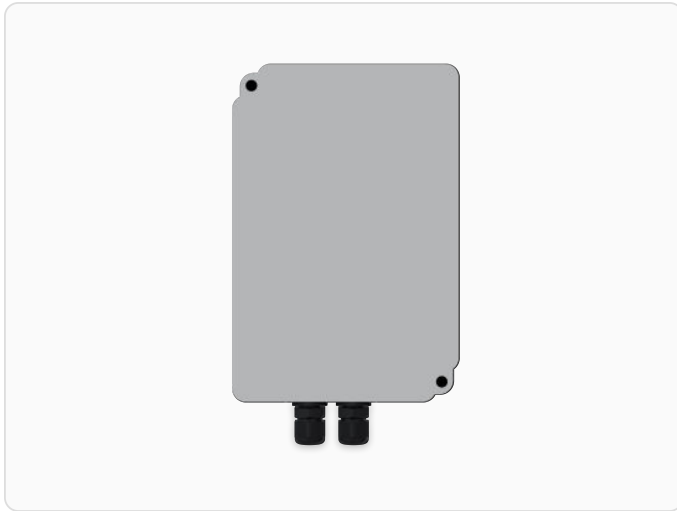
MacSet-CX1 is a robust, power-operated **4G LTE Cellular Controller** designed for reliable remote control and automation of industrial and field equipment. Equipped with **2 high-current digital outputs**, it enables secure switching of pumps, motors, valves, alarms, solenoids, lighting systems, and other electrical loads over cellular connectivity — without requiring any local Wi-Fi, Ethernet, or LoRaWAN gateway infrastructure.

Engineered for demanding outdoor and industrial environments, the MacSet-CX1 features a stable **12–24 V DC** power architecture, noise-immune circuitry, and an **IP66-rated enclosure** suitable for harsh field conditions. Its always-on cellular link ensures dependable command delivery and monitoring across farms, industrial plants, buildings, and remote infrastructure sites.

The device supports cloud integration using **MQTT / HTTPS** and can be configured locally via **BLE using the Macnman Maya App**, enabling fast installation, testing, and diagnostics. It also supports local automation logic such as timers, schedules, and fail-safe output behavior for uninterrupted operations.

Whether used for smart agriculture, industrial automation, water management, or remote asset control, **MacSet-CX1 delivers a powerful, scalable, and future-ready solution for cellular-based switching and automation.**





Key Features

Dual High-Current Digital Outputs

Enables remote switching of pumps, motors, solenoids, alarms, and lighting loads.

Industrial 12–24 V Power Operation

Stable, noise-immune power input designed for factories and outdoor field installations.

Local Automation Logic

Execute actions based on triggers, timers, or inputs — even without cloud dependency.

BLE Configuration via Maya App

Quick on-site setup, testing, and diagnostics using Bluetooth Low Energy.

Event-Based Actuation

Trigger outputs based on cloud commands, schedules, or device-side automation rules.

Firmware OTA-Ready

Future upgrades, improvements, and feature releases supported over BLE.

4G LTE Cellular Controller

Always-on remote control over cellular network without Wi-Fi, Ethernet, or LoRaWAN gateway dependency.

Outdoor IP66 Industrial-Grade Enclosure

Rugged design suitable for outdoor, dusty, and moisture-prone environments.

Cloud Ready (MQTT / HTTPS)

Integrates easily with AWS IoT Core, Azure IoT Hub, ThingsBoard, Ubidots, and custom cloud platforms.

Secure Remote Switching

Supports encrypted communication and reliable command delivery for critical operations.

Fail-Safe Output Handling

Supports power-restore behavior, default states, and safe automation routines.

Designed for Remote Industrial Control

Ideal for irrigation systems, pump automation, industrial utilities, and remote site switching.

Suited Applications

Smart Agriculture & Irrigation

Remotely control irrigation pumps, valves, solenoids, and fertigation systems using cellular connectivity.

Water & Wastewater Management

Operate pumps, blowers, aerators, chlorination units, and valve actuators at remote water sites.

Industrial Automation

Switch motors, relays, contactors, alarms, and machinery as part of factory automation workflows.

Solar & Energy Systems

Remotely activate inverter controls, battery bank relays, and system bypass/switching mechanisms.

Cold Storage & Food Industry

Switch compressors, fans, alarms, and standby systems for temperature-sensitive operations.

Smart City Deployments

Operate sirens, public lighting, fountains, signage, and remote utility controllers over cellular network.

Smart Buildings & Facilities

Control lighting circuits, exhaust fans, access gates, HVAC auxiliary systems, and building utilities.

Remote Infrastructure Control

Manage field devices such as telecom tower equipment, remote panels, and unmanned site actuators.

Pipeline & Industrial Utility Networks

Control valves, pumps, pressure release actuators, and emergency shutoff systems remotely.

Environmental Monitoring Systems

Trigger fans, blowers, alarms, or auxiliary devices based on sensor-driven automation rules.

Technical Specifications

General Specifications

Part Order Number	MacSet-CX1
Product Brand	Macnman Technologies Pvt Ltd

Interfaces Available

High Current Relay Outputs	x2 @16 Amps
RS485 Interface	1 (Switchable)
4-20 mA Analog Channels	x 2 (Switchable)
0-10 V Analog Channels	x 2 (Switchable)
Digital Inputs	x 2 (Switchable)
ADC Resolution	12-bit

Note

The device use same channels for the RS485 & Analog inputs, either can be used at a time.

Wireless Transmission

Communication Module	Quectel EC200 Series
Network Technology	4G LTE (Cat 1) 2G GSM (Fallback)
Supported LTE Bands*	B1 / B3 / B5 / B8 / B40 / B41 (Region dependent)

Supported GSM Bands*	900 / 1800 MHz
Tx Power	LTE: Up to +23 dBm GSM: Up to +33 dBm
Receiver Sensitivity*	LTE: Up to -100 dBm (typical) GSM: Up to -109 dBm (typical)
Cloud Protocols	MQTT / HTTPS
Antenna	External high-efficiency cellular antenna
SIM Support	Nano SIM
Data Transmission	Configurable reporting interval with event-based alerts

LED Status

LED Type	1 Internal Multi-Color LED
Solid Blue	Device is connected with Maya App
Flashing Blue	OTA (Over-the-Air) update in progress
Slow Blinking Green	Attempting to connect to the network
Single Red Blink	A confirmed parameter of MacSet-CX1 is being modified

Device Configuration

Configuration	Via Macnman Maya Android App
Uplink Modes	Time Based / Threshold Based / Sampling Based
Remote Configuration	JSON Format Data - from server

Supported Cloud Platforms

Supports Major Cloud Platforms	AWS IoT Core Microsoft Azure IoT Hub ThingsBoard Ubidots Datacake TagoIO Blynk Custom Cloud via MQTT / HTTPS
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Physical Characteristics

Power Supply	9-36 V DC
Operating Temperature	-40°C to +80°C
Humidity Range	0% to 100% RH (non-condensing)

Ingress Protection	IP65 standard (IP67 optional variant)
Housing Material	Glass field Nylon
Color	Matte White
Weight	~110g
Mounting Options	Pole Mounting / Wall Mounting

Electrical Characteristics

Input Voltage Range	9-36 V DC (External Supply)
Power Consumption	~100 µA in sleep mode ~50 - 250 mA during transmission
Input Impedance	>10 MΩ for voltage inputs

Communication Options

BLE Connectivity	For local configuration and diagnostics via Maya App
Cloud Protocols	MQTT / HTTPS
Serial Interface	RS485 Modbus RTU support (optional)

Certifications & Compliance

WPC	Compliant
RoHS	Conforms to RoHS directive

Security & Encryption

Data Encryption	AES-128 end-to-end encryption
Authentication	OTAA activation modes
Firmware Protection	Secure boot and signed OTA updates

MacSet CX1 Variants

VARIANT NAME	POWER SOURCE	INPUT VOLTAGE	RELAY OUTPUTS	SOLAR SUPPORT	EXTERNAL FLASH	IDEAL FOR
MacSet CX1 - PO	External Power	12-24 V DC	2			Outdoor and industrial switching applications with always-on cellular connectivity

What's in the Box?

Make sure box contains following items before installations



MacSet



Mounting Clip*



User manual & Warranty Card



IP Extender*



Mounting Screws

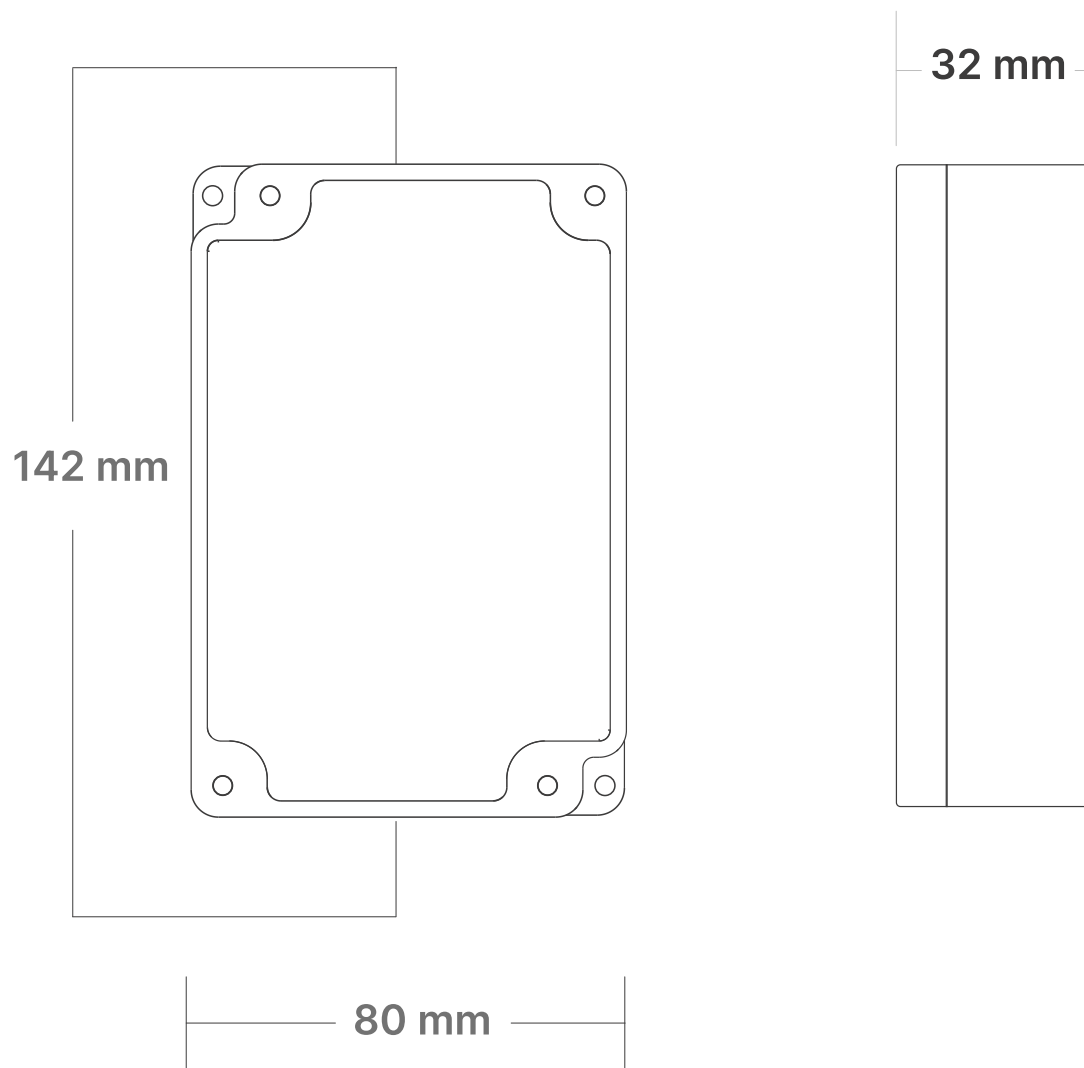


Power Adapters*

Note

Accessories must be purchased separately & may not be included in the box.

MacSet CX1 Dimensions



Frequently Asked Questions

What is the MacSet CX1 used for?

MacSet CX1 is a power-operated 4G LTE cellular relay controller for reliable remote control and automation of pumps, motors and field equipment. Typical applications include Smart Agriculture & Irrigation, Water & Wastewater Management, Industrial Automation, Smart Buildings & Facilities, Solar & Energy Systems and Remote Infrastructure Control.

What inputs and outputs does the MacSet CX1 have?

The MacSet CX1 provides the following interfaces:

- High Current Relay Outputs: x2 @16 Amps
- RS485 Interface: 1 (Switchable)
- 4-20 mA Analog Channels: x 2 (Switchable)
- 0-10 V Analog Channels: x 2 (Switchable)
- Digital Inputs: x 2 (Switchable)
- ADC Resolution: 12-bit

Which cellular networks does the MacSet CX1 support?

The MacSet CX1 uses a Quectel EC200 Series module for 4G LTE (Cat 1), 2G GSM (Fallback) connectivity. Supported LTE bands: B1 / B3 / B5 / B8 / B40 / B41 (Region dependent). Supported GSM bands: 900 / 1800 MHz. It takes a Nano SIM and sends data over MQTT / HTTPS.

How is the MacSet CX1 powered?

The MacSet CX1 is powered from 9-36 V DC (External Supply).

What is the IP rating and operating temperature of the MacSet CX1?

The MacSet CX1 has an ingress protection rating of IP65 standard (IP67 optional variant) and an operating temperature range of -40°C to +80°C. Operating humidity: 0% to 100% RH (non-condensing). Mounting options: Pole Mounting / Wall Mounting.

How is the MacSet CX1 configured?

The MacSet CX1 is configured with the Macnman Maya Android app. Supported uplink modes: Time Based / Threshold Based / Sampling Based. It can also be configured remotely with JSON data sent from the server.

Related Products and Resources

- Related datasheets: [MacSet LX1 industrial LoRaWAN relay controller](#), [MacSync CX1 4G LTE cellular data logger](#), [MacEdge CX08 industrial edge gateway](#), [MacEdge CX10 industrial edge gateway](#)
- Help: [Warranty terms and conditions](#) and [contact Macnman support](#)